PAVEL ZOSIM

TECHNICAL ARTIST | REAL-TIME SYSTEMS / SHADERS / GPU WORKFLOWS

Houdini · Unreal · Unity · HLSL · Python · C# · Compute Shaders

Demo reel & breakdowns: www.pavelzosim.com

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SUMMARY

Technical Artist specializing in procedural systems, real-time XR, and GPU-driven workflows (Houdini, HLSL, Unity, Unreal). Consistent approach across 6+ years: identify production bottleneck, architect a tool or system, eliminate it — often as the only technical generalist on small cross-disciplinary teams, working directly with art and engineering.

Experience includes GPU-driven VFX (HLSL, compute shaders), data-driven tooling (Python/C#), and performance-critical workflows for VR/XR applications. Strong background in Houdini-based procedural systems and real-time integration, with emphasis on scalability, optimization, and pipeline reliability.

Generalist by intent: shaders, procedural systems, tooling and engine integration held by one person — the shape small and mid-size teams actually need. Where there is a gap, I take the problem, study it, and solve it; that is how Houdini, HLSL and USD each entered the toolkit. On scale, I treat pipelines as contracts rather than cleverness: registries, naming conventions, data-driven inputs and written documentation, so a system stays predictable as a team grows and keeps working when its author is not in the room.

CORE SKILLS

- HLSL, Compute Shaders, Shader Graph, Unreal Materials
- Real-time VFX systems (particles, destruction)
- Parameter-driven procedural VFX systems — effects exposed as tunable parameter sets so variants are generated rather than re-authored
- PBR material authoring, MaterialX portable shading graphs, ACES colour pipeline
- GPU-driven workflows and shader integration
- Houdini (Houdini Digital Assets), procedural geometry, data-driven setups
- USD workflows, asset processing, pipeline logic
- SpeedTree for vegetation; Substance Designer and Painter, ZBrush for asset authoring
- Pipeline scripting in Python, VEX and C# — Houdini automation, Unity editor tools, exporters
- Asset validation, exporters, workflow automation
- USD workflows in Houdini — procedural scene pipelines, asset structure, cross-DCC collaboration (personal RnD)
- Git, Perforce, cross-team workflows
- Asset handoff, CI-aware production flows
- Unity, Unreal Engine 5, and proprietary in-house engines — adapting authored systems to whatever runtime and toolchain a studio already has
- VR/XR optimization (90 FPS constraints)

KEY SYSTEMS ARCHITECTED

- Anisotropic Kuwahara oil-paint filter as a custom Unreal render pass — a multi-stage compute pipeline driven by scene depth, normals and CustomStencil
- Developing a systems-driven game environment on a hybrid model — procedural world building and data-driven level construction for the bulk of the content, with deliberate manual placement where art direction demands it; the tooling exists to keep both under one workflow

- Built Houdini-to-Unreal workflows for procedural generation, structural placement, modular buildings, environment assembly, and reusable Blueprint/HISM systems
- Developed shaders, Niagara VFX, volumetric clouds, winter atmosphere, lighting, and material variation for a large-scale real-time environment
- Designed for deterministic generation, explicit art direction, and reduced manual placement while keeping artist control
- Modular HDA framework for deterministic procedural level generation with Unreal Engine integration
- Room-aware DFS topology, structural wall/ground/ceiling abstractions, asset registries, gameplay placement, procedural scatter, and manual overrides
- Separate registry and placement contracts for instanced structural modules, Blueprint actors, gameplay objects, and environmental props
- Deterministic anchor-based placement and weighted variation, with reproducible output and artist-controlled overrides
- Architected for reusable Unreal asset bindings and future data-driven / PCG workflows
- USD scene assembly in Houdini Solaris — layered scene structure so several artists can work on the same environment in parallel instead of serializing on one file
- Cross-DCC handoff exercised across Houdini, Omniverse and Blender: asset structure and referencing rather than hand-carrying files between packages
- Written up as a public breakdown on the blog, with the scene structure and layering rules documented
- Cubemap Assembly Tool — assembles six loose faces into DirectX-ordered cubemaps: automatic face detection from filenames, per-face rotate/flip correction, live 3D cube and inside-360° preview, export to cross/strip PNG/TGA/BMP layouts or DDS through DirectXTex texassemble
- Sprite Sheet Assembly Tool — ordered frame sequences to engine-ready atlases: drag reordering, animation preview, POT grid recommendations ranked by texture memory utilization, exact-grid vs POT-rescale modes with explicit distortion warnings, layout metadata encoded in the filename
- Both are pywebview/Pillow desktop apps with a service-layer architecture, isolated .venv installers for Windows and Linux, and unit tests over the core algorithms
- Procedural Tire Generator — physically accurate tire system driven by real-world parameters (width mm, aspect ratio %, rim diameter inches); custom polygon splitting for uniform tread extrusion, dual UV setup (pattern-based and cylindrical), SVG-based sidewall branding; scales consistently across any tire spec without manual correction
- Independent studios and technical artists run these in production: Procedural Skeleton / Rig Bones Generator (game-convention joint naming, APEX-ready, FBX for Unity and Unreal), Folder-Based File Importer, Material Path Manager
- Each ships with documentation and implementation notes rather than a bare download — built to survive in someone else's pipeline, not only my own
- Input: screen meshes with varied orientations and aspect ratios across multiple display configurations
- Auto-detected and corrected mesh orientation; proportionally scaled each UI element into a unified UV grid
- Output: single texture atlas covering all screens — one draw call regardless of screen count, eliminating per-screen material overhead
- Fully non-destructive and rerunnable — adding a new screen variant required zero manual UV work
- Structured export pipeline for simulation and procedural animation data — designed for predictability and maintainability over ad-hoc scripts
- Quaternion-to-Euler conversion with gimbal lock handling — stable camera and object transfer from complex 3D paths
- Smart floating-point truncation — high-fidelity data transfer without file size bloat
- Batch multi-subject export: cameras + arbitrary point clouds (VEX-driven and solver-based motion) in a single pass
- Companion AE script reconstructs data cleanly on the compositing side — no manual keyframe work

EXPERIENCE

Technical Artist | VFX Developer · Freelance Contract

April 2024 – Present

- Developed custom HLSL shaders and real-time VFX for XR cinematic experience — Blumhouse horror IP (M3GAN, The Black Phone) reimagined as immersive spatial experiences on Meta Quest 3 / 3S
- Balanced low-resolution headset rendering with streamed 4K video playback (HISPlayer) and Dolby Atmos spatial audio under varying network conditions on standalone hardware
- Prototyped effect workflows in Houdini before real-time implementation — reduced iteration cycles and ensured GPU budget compliance prior to engine integration
- Reduced shader complexity and overdraw, improving GPU frame time by ~2–4 ms on Quest 3S · premiered at Meta Connect · released publicly on Meta Quest Store
- Real-time VFX for a live mobile match-3 / casino title in Unity — boosters, blockers, power-ups, destruction, celebration and transition effects (VFX Graph, custom HLSL flipbook glow shader)
- Technical challenge: 35 booster combinations plus base effects meant ~90 separate assets, and Unity VFX Graph cannot save emitters independently the way Unreal can — so every shared element had to be duplicated per combination and every art change re-applied by hand
- Solved it with a parametric setup: integer-driven presets in shared SubGraphs (BoosterAct / ExplodeAct constants) switch emitter states inside one container, and a single root controller reads a ComboIndex from the data table and drives a pooled effect set
- Result: asset count cut from ~90 files to ~40, and a fix to a base effect is now made in one place instead of being repeated across roughly three duplicated assets — every combination that uses it picks the change up automatically
- Wrote the integration, setup and debug documentation and the prefab catalog the team works from, and set the naming and folder conventions the team now works from across shaders, graphs, subgraphs, flipbooks and prefabs

Technical Artist | VFX Artist | Motion Designer · Bully! Entertainment

June 2018 – April 2024

Small studio, wide remit — on these projects I covered concept, modeling, animation, shaders, tooling and final delivery end to end, rather than a single specialised slot.

- Served as Art Director, UI/UX, and Technical Artist — owned visual and technical decisions across app, website, graphic novel, and video formats for NASA's Artemis program narrative platform; primary bridge between art and engineering
- Architected automated Houdini pipeline generating unified texture atlas from multiple screen meshes with varying aspect ratios — shader-driven UV offset control enabling flexible element ranking with zero manual UV work per variant
- Developed UI shaders operating consistently across screen-space and world-space contexts; built procedural facial expression animation system for robot assistant (eye and eyelid movement producing expressive emotional states efficiently)
- Optimized for mobile iOS/iPad hardware constraints — reduced draw calls and shader overdraw across multi-format delivery
- 1.5M website page views · 325K+ app downloads · 350K+ graphic novel copies distributed worldwide · astronaut Frank Rubio read the content aboard the ISS · award-winning educational project
- Served as Art Director, Technical Artist, and UI/UX — owned full UI delivery pipeline across screen-space and world-space contexts; art directed within strict brand guidelines across 3 major iterations (Virtual Factory 3.0)
- Engineered internal texture export and sprite sheet generation tooling as the team’s technical generalist, working directly with the art and engineering leads — built in parallel with delivery, adopted across the team, eliminating ~30–40% of manual asset preparation overhead per cycle; tools reused in subsequent projects
- Built scalable UI pipeline processing 100+ assets via texture atlases and 9-slice sprite sheets — maintained visual consistency across screen formats, spatial UI, and physical AR floor markers readable from multiple angles
- 500+ captured leads at live exhibition · American Advertising Awards recognition · associated comic book release · client returned for expanded version
- Real-time VFX and lighting for a camera-tracked installation where visitors steered a Banshee in flight by hand movement — skeletal tracking driving gameplay and effects with no controller
- Served as Art Director and Technical Artist — directed visual language, spatial layout, and technical implementation for airport operations interactive screen
- Engineered procedural city and airfield generation using Houdini + OpenStreetMap data + Python — accurate spatial representation at scale without manual modeling

- Built timeline-driven Unity HDRP system synchronizing aircraft animation (taxi, takeoff, flight, landing) with video playback and touch-screen camera control
- Designed for 24/7 continuous operation — visual pacing, stability, and animation timing tuned to prevent viewer fatigue over extended display cycles
- Delivered as internal R&D proof-of-concept for aviation-focused interactive systems; successfully validated for long-term stability

EARLY CAREER FOUNDATION

UI/UX, Motion & 3D Designer · App, web, motion, print and photography

2005 – 2018

- Thirteen years of visual design across app and web UI/UX, motion graphics, 3D, print and photography — the foundation behind readable UI, composition, and colour judgement in later technical work
- Architected a fully procedural simulation and rendering system driven by Python — procedural camera rigs, character locomotion, crowd simulation (30+ agents), and shader-driven randomized texture generation; a complete unique sequence produced from a single script execution with no manual intervention
- Built automation tooling to systematize repetitive production tasks — the consistent pattern across the career: identify a repeating workflow, build a tool to remove the manual overhead

EDUCATION & LANGUAGES

ULIM – Universitatea Liberă Internațională din Moldova · Chisinau, Moldova

2005 – 2018

Computer Science and CG Design (self-directed) — advanced self-learning in Houdini, Unity/Unreal, Python for procedural workflows and tool development.

Russian — Native

English — Professional Working Proficiency